

# **Enterprise Digital Infrastructure**

**Master Degree in Computer Engineering**

**Maria Carla Calzarossa**

**Spring 2026**

© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

1

## **Outline**

Syllabus

Resources

Schedule of the classes

Rules

Exam and scoring

© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

2

# Syllabus

Enterprise Digital Infrastructure (EDI) under two perspectives:

- Methodological
- Technological

Topics:

- Definitions & basic concepts
- Performance monitoring
- *Performance evaluation techniques*
- Networked applications and protocols
  - Naming services: DNS
  - Web services: HTTP
  - Email services: SMTP, POP3, IMAP4
  - *Other services: video streaming, file sharing, ...*

© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

3

## What you will learn

Hard skills

- Methodological, technical and practical competencies on digital infrastructures

Soft skills

- Technical writing skills
- Team work and communication/presentation skills

Prerequisites

- Basic concepts of computer networks
- Linux operating system

© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

4

## Resources

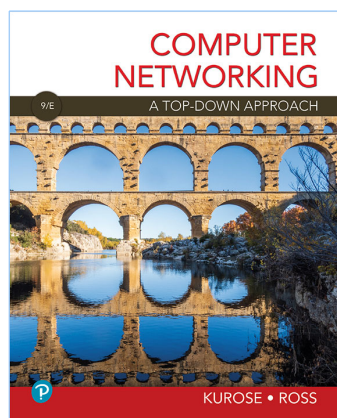
- Books and technical reports
- Lecture slides
  - PDF available on Google Drive **after** each lecture
- Lecture notes

© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

5

## Resources: reference book



Chapter 2 Application Layer

Chapter 8 Security in Computer Networks

© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

6

## Other resources: RFCs

### RFC Editor

#### The Series

Document Retrieval  
Errata  
FAQ  
Format Change FAQ  
History  
About Us  
Other Information

#### For Authors

Publication Process  
Publication Queue  
Style Guide  
I-D Author Resources  
Independent Submissions

#### Mailing Lists

rfc-dist@rfc-editor.org is for RFC publication announcements. rfc-interest@rfc-editor.org is for discussion of the RFC series and related functions.

#### Sponsor



#### The RFC Series

The RFC Series (ISSN 2070-1721) contains technical and organizational documents about the Internet, including the specifications and policy documents produced by five streams: the Internet Engineering Task Force (IETF), the Internet Research Task Force (IRTF), the Internet Architecture Board (IAB), Independent Submissions, and Editorial.

#### Browse the RFC Index

HTML (ascending) • HTML (descending) • TXT • XML

Note: These files are large.

#### Browse RFCs by Status

Official Internet Protocol Standards

Best Current Practice

Informational • Experimental • Historic

Uncategorized (Early RFCs)

• • • • •

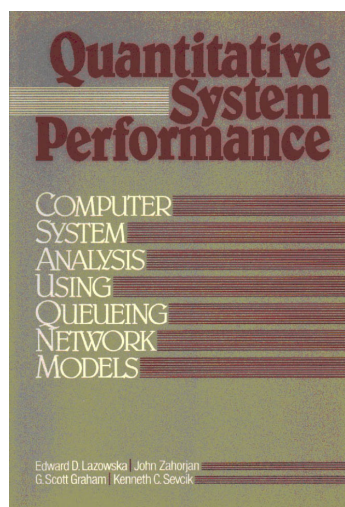
RFC Status Changes

© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

7

## Another book



#### I. Preliminaries

- 1. An Overview of Queueing Network Modelling
- 2. Conducting a Modelling Study
- 3. Fundamental Laws
- 4. Queueing Network Model Inputs and Outputs

#### II. General Analytic Techniques

- 5. Bounds on Performance
- 6. Models with One Job Class
- 7. Models with Multiple Job Classes
- 8. Flow Equivalence and Hierarchical Modelling

#### III. Representing Specific Subsystems

- 9. Memory
- 10. Disk I/O
- 11. Processors

#### IV. Parameterization

- 12. Existing Systems
- 13. Evolving Systems
- 14. Proposed Systems

#### V. Perspective

- 15. Extended Applications
- 16. Using Queueing Network Modelling Software

#### VI. Appendices

- 17. Constructing a Model from RMF Data
- 18. An Implementation of Single Class, Exact MVA
- 19. An Implementation of Multiple Class, Exact MVA
- 20. Load Dependent Service Centers

© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

8



© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

9



## House Rules

☐

Eat and drink only in the kitchen

☐

You make a mess, you clean it up

☐

Try everything on your plate

☐

No hitting, kicking, or biting

☐

Tell the truth

☐

Keep your promises

☐

Say please and thank you

☐

Work hard

## HOUSE RULES



Leave shoes at the door



Tech-free meals, no screens



One in, one out



Respect quiet resting spaces



Switch off lights when leaving



Water the plants regularly



Speak only with kindness



No dirty dishes overnight



Celebrate small wins together

## HOUSE RULES


We share & look after toys


No screaming or shouting


No jumping on the furniture


We use kind words to each other


We listen to each other


We eat nicely at the table


We ask before taking or doing


We take off our shoes inside


We use our manners

© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

10

© University of Pavia 2026

5

## Schedule

- Schedule: 12 hours per week (starting at 9:15am)
- Lectures, lab activities & seminars
  - Theory [*Slides & blackboard*]
  - Examples & exercises [*Blackboard*]
  - Hands-on sessions [*B2 lab*] **CHECK YOUR ACCOUNT**
  - Case studies [*Experts from industries*]
  - Self-assessments [*On paper*] **BRING YOUR OWN PENS**

© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

11

## Rules

- No phones during classes



- Multiple short breaks



© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

12

## Exam

The exam is subdivided in four parts

Written exam

*Oral exam*

*Practical activities with an individual report*

*Team project work*

The written exam consists of two parts:

Open and multiple-choice questions about the theory

Exercises about applications of the theory

© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

13

## Assessments & final score

### Option #1

- Written exam (up to 24 points)
- Practical activities with an individual report (up to 4 points)
- Team project work (up to 4 points)
- *Optional oral exam* (up to 2 points)

### Option #2

- Written exam (up to 20 points)
- Oral exam (up to 8 points)
- Practical activities with an individual report (up to 4 points)

### Option #3

- Written exam (up to 16 points)
- Oral exam (up to 12 points)
- Team project work (up to 4 points)

### Option #4

- Written exam (up to 14 points)
- Oral exam (up to 18 points)

**Options #1 and #2 require lab activities attendance (in-out attendance list)**

© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

14

## Timeline of the assessment

Project work presentations: scheduled towards the end of the semester

Lab report: due by the end of the semester

Written exams: scheduled in the official dates published on the Engineering School website

Oral exams: scheduled some days after the written exam

© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

15

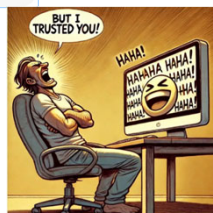
## GenAI policy

To avoid undermining the learning process and problem-solving abilities, the usage of any GenAI tool for assignments related to the class is **strictly forbidden**



### Writing is thinking

On the value of human-generated scientific writing in the age of large-language models.



© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

16



## Feedback

Your feedback are always very welcome

- Raise your hand and ask questions or provide your feedback
- Send me an email
- or ....

Anonymous surveys during the semester

© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

17

## How to reach me

- Office @F Floor, lab @D Floor (Performance Evaluation lab) both in the “Nave” building
- Email: [mcc@unipv.it](mailto:mcc@unipv.it)
- Office hours: *in person* or *online* by appointment (email using your **official** email address)

© M. C. Calzarossa 2026

Enterprise Digital Infrastructure

18



© M. C. Calzarossa 2026

Enterprise Digital Infrastructure